

## WOODWARD FIELD, SC, USA

WMO: 720603

Lat: 34.283N

Lon: 80.567W

Elev: 92

StdP: 100.22

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00195

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |              |            |            | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|--------------|------------|------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%          |            |            | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP           | HR         | MCDB       | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-5.1 | (c)<br>-2.7 | (d)<br>-15.3                  | (e)<br>1.0 | (f)<br>-0.7 | (g)<br>-12.3 | (h)<br>1.3 | (i)<br>1.2 | (j)<br>9.1            | (k)<br>11.2 | (l)<br>8.1 | (m)<br>10.3 | (n)<br>0.8            | (o)<br>290 | (p)<br>0.374 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.9                   | 36.3            | 25.0 | 35.0 | 24.5 | 33.7 | 24.2 | 27.4                | 31.8 | 26.4 | 31.3 | 25.7 | 30.9 | 3.2                  | 230  |

| Dehumidification DP/MCDB and HR |             |             |             |             |             |             |             |             | Enthalpy/MCDB |             |             |             |             |             | Extreme<br>Max WB |
|---------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------|-------------|-------------|-------------|-------------|-------------|-------------------|
| 0.4%                            |             |             | 1%          |             |             | 2%          |             |             | 0.4%          |             | 1%          |             | 2%          |             |                   |
| DP                              | HR          | MCDB        | DP          | HR          | MCDB        | DP          | HR          | MCDB        | Enth          | MCDB        | Enth        | MCDB        | Enth        | MCDB        |                   |
| (a)<br>26.3                     | (b)<br>22.0 | (c)<br>29.3 | (d)<br>25.2 | (e)<br>20.5 | (f)<br>28.0 | (g)<br>24.1 | (h)<br>19.2 | (i)<br>27.4 | (j)<br>87.4   | (k)<br>32.1 | (l)<br>82.5 | (m)<br>31.2 | (n)<br>79.2 | (o)<br>31.9 | (p)<br>33.7       |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 7.7               | 6.6  | 5.5 | -8.9                       | 38.3 | 2.6                | 2.3 | -10.8                                              | 40.0 | -12.3      | 41.3 | -13.8      | 42.6 | -15.7      | 44.3 |
|                   |      |     | -9.7                       | 29.1 | 2.6                | 2.8 | -11.6                                              | 31.1 | -13.1      | 32.8 | -14.6      | 34.3 | -16.5      | 36.3 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 17.9  | 6.7   | 9.2   | 12.7  | 18.2  | 22.9  | 26.4  | 27.9  | 26.7  | 24.3  | 18.1  | 11.8  | 9.5  | (6)  |
| (7)  |                                                                                  | DBStd       | 8.45  | 5.50  | 5.42  | 5.22  | 4.31  | 3.49  | 2.41  | 1.95  | 2.30  | 3.03  | 4.38  | 4.63  | 4.76 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 347   | 133   | 76    | 32    | 2     | 0     | 0     | 0     | 0     | 0     | 1     | 33    | 71   | (8)  |
| (9)  |                                                                                  | HDD18.3     | 1406  | 361   | 259   | 185   | 54    | 7     | 0     | 0     | 0     | 2     | 58    | 202   | 278  | (9)  |
| (10) |                                                                                  | CDD10.0     | 3235  | 32    | 53    | 117   | 247   | 401   | 491   | 555   | 519   | 429   | 252   | 86    | 54   | (10) |
| (11) |                                                                                  | CDD18.3     | 1251  | 1     | 3     | 12    | 50    | 149   | 241   | 297   | 261   | 181   | 51    | 4     | 3    | (11) |
| (12) |                                                                                  | CDH23.3     | 12365 | 4     | 22    | 127   | 516   | 1412  | 2511  | 3167  | 2543  | 1593  | 429   | 36    | 6    | (12) |
| (13) |                                                                                  | CDH26.7     | 5539  | 0     | 1     | 21    | 146   | 581   | 1214  | 1579  | 1186  | 686   | 117   | 9     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.0   | 1.9   | 2.2   | 2.3   | 2.3   | 2.1   | 1.9   | 1.6   | 1.9   | 1.8   | 1.8   | 1.8   | (14) |      |
| (15) | Precipitation                                                                    | PrecAvg     | 1164  | 113   | 91    | 110   | 74    | 76    | 112   | 128   | 122   | 94    | 80    | 75    | 82   | (15) |
| (16) |                                                                                  | PrecMax     | 1565  | 229   | 190   | 258   | 167   | 184   | 261   | 252   | 274   | 224   | 430   | 217   | 204  | (16) |
| (17) |                                                                                  | PrecMin     | 778   | 18    | 21    | 10    | 4     | 16    | 16    | 11    | 14    | 6     | 0     | 8     | 10   | (17) |
| (18) |                                                                                  | PrecStd     | 185   | 51    | 46    | 51    | 44    | 42    | 65    | 70    | 62    | 57    | 78    | 45    | 45   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 23.7  | 26.1  | 28.8  | 32.3  | 36.2  | 37.6  | 37.9  | 38.8  | 36.0  | 32.4  | 27.1  | 24.0 | (19) |
| (20) |                                                                                  |             | MCWB  | 17.7  | 18.2  | 18.3  | 20.9  | 23.7  | 24.1  | 25.6  | 28.4  | 23.4  | 23.6  | 20.8  | 21.3 | (20) |
| (21) |                                                                                  | 2%          | DB    | 20.2  | 23.0  | 27.1  | 30.0  | 33.7  | 35.9  | 36.4  | 35.1  | 33.8  | 29.1  | 23.7  | 22.2 | (21) |
| (22) |                                                                                  |             | MCWB  | 15.5  | 16.6  | 17.5  | 19.9  | 23.0  | 24.3  | 25.4  | 25.2  | 22.9  | 21.8  | 17.6  | 19.4 | (22) |
| (23) |                                                                                  | 5%          | DB    | 17.9  | 21.2  | 24.1  | 28.0  | 32.1  | 33.9  | 35.0  | 33.6  | 32.5  | 27.6  | 22.1  | 20.0 | (23) |
| (24) |                                                                                  |             | MCWB  | 14.6  | 15.6  | 16.3  | 19.5  | 22.4  | 23.9  | 25.0  | 24.8  | 22.9  | 20.8  | 17.1  | 17.7 | (24) |
| (25) |                                                                                  | 10%         | DB    | 16.2  | 18.7  | 22.3  | 26.9  | 30.0  | 32.6  | 33.1  | 32.4  | 31.0  | 25.9  | 19.8  | 17.8 | (25) |
| (26) |                                                                                  |             | MCWB  | 13.0  | 14.3  | 15.1  | 18.8  | 21.8  | 23.7  | 24.7  | 24.5  | 22.9  | 20.2  | 16.4  | 15.5 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 19.0  | 19.5  | 19.5  | 24.2  | 27.6  | 26.3  | 28.0  | 30.1  | 26.0  | 25.5  | 22.1  | 22.5 | (27) |
| (28) |                                                                                  |             | MCD B | 21.3  | 23.1  | 25.1  | 26.5  | 30.5  | 33.7  | 32.9  | 35.3  | 32.1  | 28.4  | 24.2  | 22.8 | (28) |
| (29) |                                                                                  | 2%          | WB    | 17.6  | 18.4  | 18.7  | 22.5  | 25.2  | 25.7  | 27.2  | 27.8  | 25.0  | 24.3  | 20.2  | 20.4 | (29) |
| (30) |                                                                                  |             | MCD B | 19.1  | 22.0  | 24.3  | 27.0  | 30.6  | 32.5  | 32.3  | 32.0  | 30.3  | 27.2  | 22.1  | 21.5 | (30) |
| (31) |                                                                                  | 5%          | WB    | 15.9  | 17.0  | 17.9  | 21.3  | 23.8  | 25.1  | 26.3  | 26.5  | 24.3  | 23.2  | 18.6  | 18.0 | (31) |
| (32) |                                                                                  |             | MCD B | 17.6  | 20.0  | 22.3  | 25.6  | 28.8  | 31.5  | 31.8  | 30.3  | 29.2  | 25.8  | 20.4  | 18.7 | (32) |
| (33) |                                                                                  | 10%         | WB    | 13.3  | 15.1  | 16.9  | 20.0  | 22.8  | 24.5  | 25.7  | 25.6  | 23.6  | 21.6  | 17.2  | 16.0 | (33) |
| (34) |                                                                                  |             | MCD B | 14.9  | 17.9  | 20.4  | 24.7  | 27.7  | 30.4  | 31.1  | 30.0  | 28.1  | 24.6  | 18.8  | 17.9 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 11.4  | 11.8  | 12.7  | 13.1  | 11.8  | 11.3  | 10.9  | 10.2  | 10.9  | 12.1  | 12.3  | 10.7 | (35) |
| (36) |                                                                                  |             | MCDBR | 13.8  | 13.7  | 15.2  | 14.7  | 14.4  | 12.9  | 12.6  | 11.8  | 13.3  | 13.4  | 14.4  | 11.6 | (36) |
| (37) |                                                                                  |             | MCWBR | 9.3   | 8.5   | 7.3   | 6.5   | 5.5   | 4.1   | 3.6   | 3.8   | 4.3   | 6.3   | 9.5   | 8.5  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 11.2  | 11.9  | 12.7  | 11.8  | 11.5  | 11.4  | 10.8  | 10.4  | 11.2  | 9.4   | 11.9  | 9.5  | (38) |
| (39) |                                                                                  |             | MCWBR | 9.2   | 8.6   | 6.8   | 6.2   | 5.1   | 4.1   | 4.0   | 4.3   | 4.1   | 5.3   | 9.5   | 8.4  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.319 | 0.329 | 0.365 | 0.412 | 0.473 | 0.502 | 0.552 | 0.522 | 0.440 | 0.395 | 0.343 | 0.316 | (40) |      |
| (41) |                                                                                  | taud        | 2.499 | 2.467 | 2.395 | 2.252 | 2.133 | 2.093 | 1.975 | 2.047 | 2.263 | 2.360 | 2.463 | 2.545 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 891   | 919   | 908   | 875   | 823   | 796   | 753   | 769   | 821   | 832   | 853   | 876   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 84    | 97    | 113   | 136   | 156   | 162   | 181   | 166   | 128   | 107   | 87    | 77    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.75  | 3.38  | 4.47  | 5.61  | 6.13  | 6.41  | 6.07  | 5.52  | 4.75  | 3.95  | 3.02  | 2.40  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.14  | 0.36  | 0.33  | 0.41  | 0.49  | 0.45  | 0.37  | 0.26  | 0.40  | 0.48  | 0.24  | 0.22  | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (68 neighbors) |  | +0.54 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +180    | +130    |

Nomenclature: See separate page